

TrES-2b

R-1283

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_130513 · created 2026-07-31T15:18:30+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456425.77863 +/- 0.00022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.015
Transit depth (Rp/Rs)^2	1.95 +/- 0.42 [%]
Orbital Inclination (inc)	83.94 +/- 0.8
Airmass coefficient 1 (a1)	1.24 +/- 0.013
Airmass coefficient 2 (a2)	-0.159 +/- 0.0037
Scatter in the residuals of the lightcurve fit is	0.98 %
Best Comparison Star	None
Optimal Aperture	4.98
Optimal Annulus	7.28
Transit Duration (day)	0.076 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.015 vs 0.1254 (deviation 0.91σ)
Duration fit vs published	0.076 d vs 0.0746 d (deviation 0.11σ)
Mid-time O–C	0.5 min
Depth SNR	9.3
Residual scatter	0.98 %

Light curve

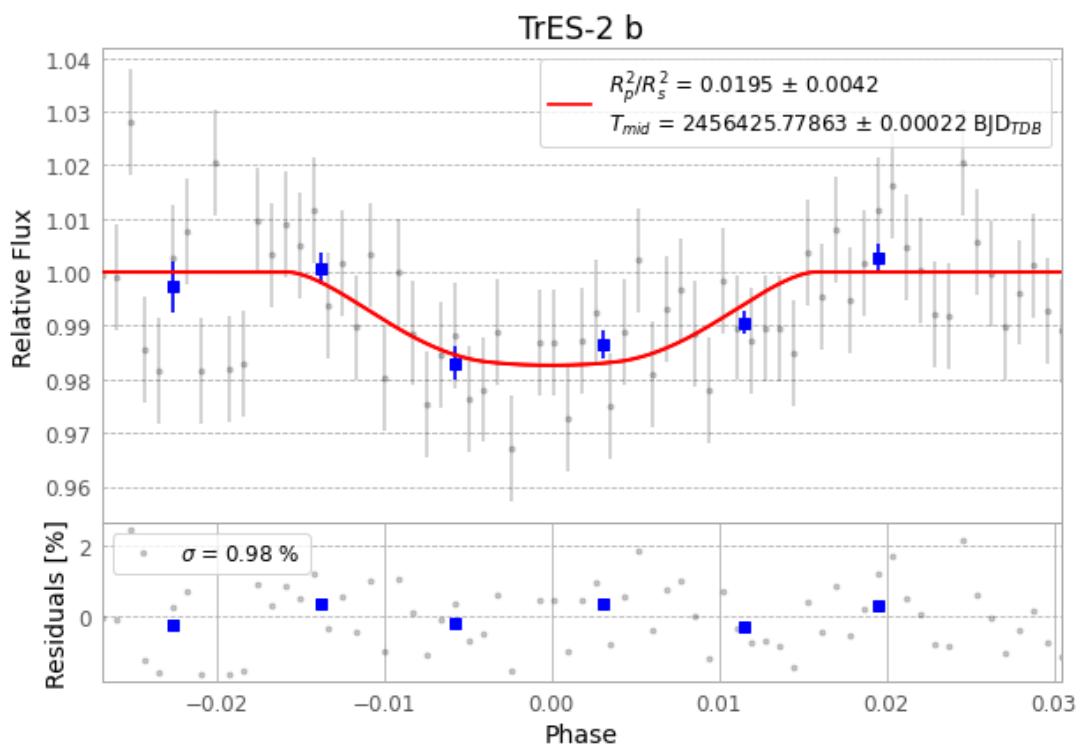


Figure 1 — FinalLightCurve_TrES-2 b_2013-05-12.png (SHA-256 49466c4be34eebbd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [269, 130], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a7278c2cd812e5b2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), TRES-2130513050313.FITS → TRES-2130513082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-130513.log (b686941484ac...), FinalLightCurve_TrES-2 b_2013-05-12.png (49466c4be34e...), FinalLightCurve_TrES-2 b_2013-05-12.csv (8ade550ae1ba...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 15:18Z.